Particle metrics: gradients and quadratic forms on general arrangements

D. Teaney[1]

[1]*Department of Physics and Astronomy, Stony Brook University, Stony Brook, New York 11794, USA*

(Dated: November 7, 2026)

Abstract

On an irregular set of particles, gradients are obtained by fitting bond differences, and quadratic forms in the gradients by bond sums. Both are controlled by the bond moment, the push-forward of the particles' own metric to physical space. The fit is exact for linear fields on any arrangement. The bond sum is the physical quadratic form only if the bond moment is proportional to the inverse metric of the fluid, flat at rest and comoving in motion. With bond weights evaluated at the comoving distance this holds for smoothly distributed particles up to sampling corrections, which we measure and remove with a small per-particle reweighting.

I. INTRODUCTION AND OVERVIEW

A particle method replaces a continuous fluid by a finite, irregular set of points. The continuum equations, however, are written in terms of gradients of fields and of quadratic forms in those gradients, contracted with a metric. A dissipative function, for example, is the square of a strain rate. On the particles both objects must be rebuilt from *bonds*: the separations $\boldsymbol{r}_{ab}$ between a particle and its neighbours, and the differences $\Delta\phi_{ab}$ of the fields across them. This note collects the general machinery for doing so. None of it is specific to dissipation, and it applies to any arrangement of particles and to a moving fluid.

a. The idea. Two operations recur in any particle description.

1. *Gradients* are obtained by fitting the bond differences of a field with a linear field (Operation 1).

2. *Quadratic forms* in the gradients are obtained as weighted sums over bonds of products of differences (Operation 2).

Both are controlled by a single matrix per particle, the *bond moment* M_a: the weighted second moment of the bond vectors of particle a. It is the matrix that is inverted to fit a gradient, and it is the metric that a bond sum actually uses to contract two gradients. The physics, on the other hand, contracts gradients with the inverse metric of the fluid: the flat metric at rest, and the comoving metric ℓ in motion. The construction works when these two agree.

b. The storyline. The note proceeds in the following steps.

- Sect. II introduces a general spatial metric h_{ij}, which covers both the fluid at rest and the moving fluid. It measures bonds, through the comoving distance that sets the bond weights, and its inverse contracts gradients.

- Sect. III defines the bond moment. Labelling the particles by a reference lattice shows that the bond moment is the push-forward of the particles' own metric to physical space. For smoothly distributed particles its continuum value is fixed by the metric in the weights.

- Sections IV and V state the two operations. The fit is exact for linear fields on any arrangement and needs only that M_a be invertible. The bond sum splits exactly into a resolved part, contracted with M_a, and a non-negative residual. It is the physical quadratic form only if M_a is proportional to the inverse metric.

- Sect. VI absorbs n_a into normalized weights w_{ab} and reads them as faces: the fit becomes Gauss's theorem and the bond sum a graph Laplacian.
- Sect. VII explains what sets M_a: the shape of the weights, not the arrangement of the particles. The only departure is sampling error.
- Sect. VIII measures that sampling error, at rest and in motion.
- Sect. IX removes it with a small per-particle reweighting Λ_a. Sections X and XI show that Λ_a vanishes in the continuum limit and describe what the code does when it cannot be found.

c. Main results.

- The fit, eq. (18), is the chain rule through the particle map, and is exact on any arrangement and with any weights.
- The bond sum obeys the exact identity eq. (24): resolved part plus residual. For smooth fields the residual is smaller by $(h/L)^2$.
- With weights evaluated at the comoving distance, the bond moment is proportional to the inverse metric up to sampling corrections, eq. (15). The particles' arrangement, including a smooth deformation or a Lorentz contraction, does not enter.
- The sampling error falls as $(a/h)^2$ for bounded disorder and grows by about γ in motion. The correction Λ_a, eq. (41), removes it exactly.

Throughout, d_s is the number of spatial dimensions spanned by the particles ($d_s = 2$ in the figures). In the application of Ref. [1] it need not equal the physical dimension d that appears in the bulk viscosity.

d. Notation. Particles sit at positions $\boldsymbol{x}_a$ and carry entropies s_a. The smoothed entropy density is $S(\boldsymbol{x}) = \sum_b s_b W(\boldsymbol{x}-\boldsymbol{x}_b)$, with an SPH kernel W of smoothing length h and support $2h$. Each particle is assigned the volume its entropy occupies at the local entropy density,

$$V_a = \frac{s_a}{S_a}, \qquad S_a \equiv S(\boldsymbol{x}_a), \tag{1}$$

which is the SPH quadrature weight, $\sum_a V_a f(\boldsymbol{x}_a) \simeq \int f$. Bonds carry a weight ψ_{ab} that is a function of the bond length, by default $\psi = -W'/r$ with its own range h_ψ. The typical particle spacing is a. The fields ϕ are arbitrary. In the application of Ref. [1], where this machinery is used to build a one-level (bond) dissipative function, the field is $\mathsf{b} = \boldsymbol{v}/T$.

II. THE METRIC

We first fix how bonds are measured and how gradients are contracted. A single metric does both, and choosing it is the only place where the motion of the fluid enters.

All bond vectors $r^i_{ab} = x^i_b - x^i_a$ are taken at the same lab time. We measure them with a spatial metric h_{ij} attached to particle a, and contract gradients with its inverse h^{ij}, $h^{ij}h_{jk} = \delta^i{}_k$. The two cases are

$$\begin{aligned} &\text{at rest:} && h_{ij} = \delta_{ij}, && h^{ij} = \delta^{ij}, \\ &\text{moving:} && h_{ij} = (\ell^{-1})_{ij} = \delta_{ij} + \gamma^2 v_i v_j, && h^{ij} = \ell^{ij} = \delta^{ij} - v^i v^j. \end{aligned} \tag{2}$$

The metric always carries indices; a plain h is the smoothing length.

a. Lengths. The length of a bond in the metric is the comoving distance in the rest frame of a,

$$\rho_{ab}^2 = h_{ij}\, r_{ab}^i r_{ab}^j = \Delta_{\mu\nu}(u_a)\, \Delta x_{ab}^\mu \Delta x_{ab}^\nu, \qquad \Delta x_{ab}^\mu = (0, \boldsymbol{r}_{ab}). \tag{3}$$

It is a Lorentz scalar, the distance between the two equal-lab-time points as measured in the rest frame of a. For a moving fluid $\det h_{ij} = \gamma^2$, so a region of lab volume $\mathcal{V}$ has rest-frame volume $\gamma\mathcal{V}$.

b. Gradients. The inverse h^{ij} is the metric that contracts spatial gradients. Consider a field that does not change along the flow,

$$u^\mu \partial_\mu \phi = 0, \qquad \text{i.e.} \qquad \partial_t \phi = -v^i \partial_i \phi. \tag{4}$$

Its rest-frame gradient squared is

$$\Delta^{\mu\nu} \partial_\mu \phi\, \partial_\nu \phi = g^{\mu\nu} \partial_\mu \phi\, \partial_\nu \phi + (u^\mu \partial_\mu \phi)^2 = \partial_i \phi\, \partial_i \phi - (v^i \partial_i \phi)^2 = h^{ij}\, \partial_i \phi\, \partial_j \phi. \tag{5}$$

This is how ℓ^{ij} enters the dissipative function of a moving fluid, as $\mathrm{tr}(\ell S \ell S)$, and why the bond form must reproduce it.

c. Weights. The bond weights are evaluated at the comoving distance, $\psi_{ab} = \psi(\rho_{ab})$. They are round in the metric h_{ij}: the stencil $\rho_{ab} < 2h_\psi$ is a circle at rest and, for a moving fluid, an ellipse contracted by $1/\gamma$ along $\boldsymbol{v}$.

III. LABELS AND THE BOND MOMENT

With the metric in hand, we turn to the particles. To see how their arrangement enters, it helps to label them by a reference lattice and to describe their positions as a map from labels to space.

Label each particle by the coordinates of a reference lattice, $a = \sigma^I$ with $I = 1, \ldots, d_s$, for example the lattice the particles started on. In this reference configuration the positions coincide with the labels,

$$x^i = \delta^i{}_I\, \sigma^I, \qquad r_{ab}^i = \delta^i{}_I\, \Delta\sigma_{ab}^I \qquad \text{(reference lattice)}. \tag{6}$$

As the fluid moves, the positions define a map $\boldsymbol{x}(\sigma)$, with deformation gradient

$$e^i{}_I = \frac{\partial x^i}{\partial \sigma^I}, \qquad r_{ab}^i \simeq e^i{}_I\, \Delta\sigma_{ab}^I, \tag{7}$$

and $e^i{}_I = \delta^i{}_I$ on the reference lattice. Both operations are controlled by the bond moment

$$M_a^{ij} = \sum_{b \neq a} V_b\, \psi_{ab}\, r_{ab}^i r_{ab}^j. \tag{8}$$

a. The bond moment as a push-forward. The bond moment is the translation between the particles and coordinate space. The particles carry their own (inverse) metric on the labels, the moment of the label differences,

$$\mu^{IJ} = \sum_{b\neq a} V_b\, \psi_{ab}\, \Delta\sigma^I_{ab}\Delta\sigma^J_{ab}, \tag{9}$$

with $\psi_{ab} = \psi(\rho_{ab})$ the physical weight as before. The bond moment is its push-forward by the particle map,

$$M_a^{ij} = e^i{}_I\, \mu^{IJ}\, e^j{}_J, \tag{10}$$

and eq. (26) below asks that it agree with $n_a h^{ij}$. We never need to evaluate μ^{IJ}, which would require knowing each particle's place on the reference lattice; it serves only to interpret M_a.

b. The continuum moment. Suppose the particles sample space smoothly: finely on the scale of the stencil ($a \ll h$), and with volumes $V_b = s_b/S_b$, eq. (1), that compensate the local particle density. The second condition requires the density to vary slowly across the stencil ($h \ll L$), since S_b is smoothed over h. The density itself need not be constant. The bond sum is then a quadrature of an integral,

$$M_a^{ij} \simeq \int d^{d_s}r\ \psi(\rho)\, r^i r^j, \qquad \rho^2 = h_{kl} r^k r^l. \tag{11}$$

The integral is evaluated by changing variables to the coordinates in which the stencil is round, $s^i = (h^{1/2})^i{}_j r^j$, so that

$$\rho = s \equiv |\boldsymbol{s}|, \qquad d^{d_s}r = \frac{d^{d_s}s}{\sqrt{\det h_{ij}}}. \tag{12}$$

By rotational symmetry in $\boldsymbol{s}$,

$$\int d^{d_s}s\ \psi(s)\, s^k s^l = n_0\, \delta^{kl}, \tag{13}$$

and so

$$\int d^{d_s}r\ \psi(\rho)\, r^i r^j = \frac{n_0}{\sqrt{\det h_{ij}}}\, h^{ij}, \qquad n_0 = \frac{1}{d_s}\int d^{d_s}s\ \psi(s)\, s^2. \tag{14}$$

Thus, for smoothly distributed particles,

$$M_a^{ij} = n_a\, h^{ij} + \big(\text{sampling corrections}\big), \qquad n_a \simeq \frac{n_0}{\sqrt{\det h_{ij}}}. \tag{15}$$

The corrections are the sampling error of a finite, irregular star, which falls with h/a, and an $O(h^2/L^2)$ error from the smoothed volumes. They are measured in Sect. VIII and removed by Λ_a in Sect. IX. Fig. 1 shows one star at rest and in motion.

IV. OPERATION 1: GRADIENTS

We now state the two operations. The first rebuilds a gradient from bond differences. It uses the bond moment only through its inverse, and needs no metric.

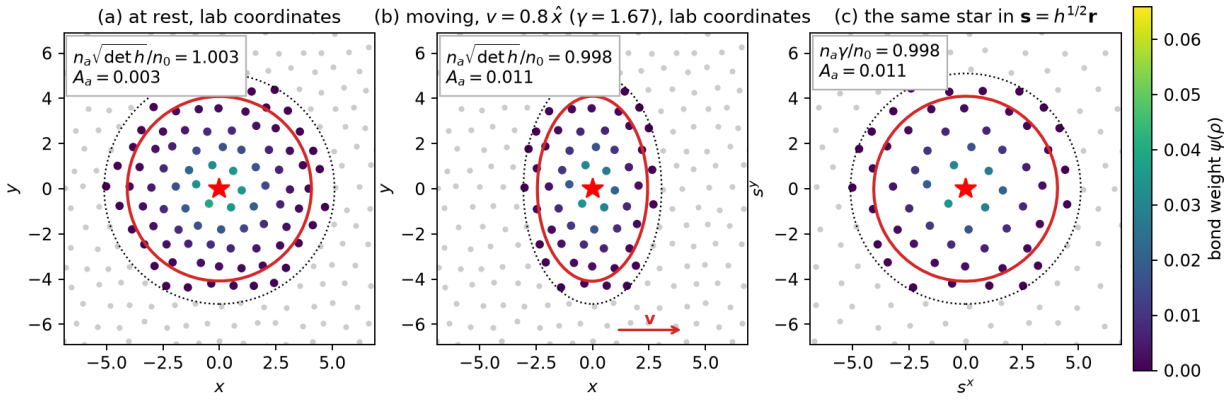


FIG. 1. One star (red star) on a jittered ($j = 0.3$) hexagonal lab lattice, $h/a = 2.55$, $\psi = -W'/r$, and V_b the lattice cell area. Dots inside the stencil $\rho < 2h$ (dotted) are coloured by $\psi(\rho)$. The red ellipse is the measured moment, $\boldsymbol{r}^T(M_a/n_a)^{-1}\boldsymbol{r} = \text{const}$. (a) At rest, $M_a \simeq n_0\mathbb{1}$. (b) Moving with $v = 0.8\,\hat{x}$ ($\gamma = 5/3$): the stencil and the moment are contracted along $\boldsymbol{v}$, and $M_a = (0.214, 0.603)$ on the diagonal, against $(n_0/\gamma)\ell = (0.216, 0.600)$. (c) The same star in $\boldsymbol{s} = h^{1/2}\boldsymbol{r}$, where the stencil is round and the moment is isotropic, $n_a\gamma/n_0 = 0.998$. The residual anisotropy A_a (Sect. VIII) is the sampling error; it is larger in motion because the contracted stencil holds $1/\gamma$ as many neighbours. Produced by `examples/stencil_metric.py`.

Let ϕ be any field carried by the particles: a component of b, a temperature, a chemical potential, and so on. Its bond differences are

$$\Delta\phi_{ab} = \phi_b - \phi_a. \tag{16}$$

The gradient at a is the linear field that best fits these differences:

$$\nabla_i\phi = \arg\min_{g_i} \sum_{b\neq a} V_b\,\psi_{ab}\big(\Delta\phi_{ab} - g_j\,r^j_{ab}\big)^2. \tag{17}$$

Setting the derivative with respect to g_i to zero gives

$$\nabla_i\phi = \tilde{\nabla}^j\phi\,(M_a^{-1})_{ji}, \qquad \tilde{\nabla}^j\phi = \sum_{b\neq a} V_b\,\psi_{ab}\,\Delta\phi_{ab}\,r^j_{ab}. \tag{18}$$

Here M_a^{ij} carries upper indices, so its inverse $(M_a^{-1})_{ij}$ carries lower ones. Here $\nabla_i\phi$ always denotes this fit, a property of the particle; we write $(\nabla_i\phi)_a$ when the particle must be explicit, and $\partial_i\phi$ for the true derivative. For the components of b, $\phi = \beta_k$, the fit is the matrix $(\nabla\mathsf{b})_a$ with components $\nabla_i\beta_k$ of eq. (40) of Ref. [1]. The fit involves no metric: the metric enters only through the weights.

Now suppose that ϕ is linear in the labels near a, so that

$$\Delta\phi_{ab} = \frac{\partial\phi}{\partial\sigma^J}\,\Delta\sigma^J_{ab}. \tag{19}$$

With the label moment μ^{JK} of eq. (9), $\tilde{\nabla}^i\phi$ and the bond moment become

$$\tilde{\nabla}^i\phi = \frac{\partial\phi}{\partial\sigma^J}\,\mu^{JK}\,e^i{}_K, \qquad M_a^{ij} = e^i{}_J\,\mu^{JK}\,e^j{}_K. \tag{20}$$

This gives the interpretation of $\tilde{\nabla}^i\phi$: it is the label gradient of ϕ with its index raised by μ^{JK} and pushed forward to physical space, i.e. the gradient with its index raised by the bond moment, $\tilde{\nabla}^i\phi = M_a^{ij}\,\partial_j\phi$. The fit lowers the index again with M_a^{-1}. Inserting $\tilde{\nabla}^i\phi$ and M_a^{ij} into eq. (18), the factors of μ cancel:

$$\nabla_i\phi = \frac{\partial\phi}{\partial\sigma^J}\,(e^{-1})^J{}_i = \frac{\partial\phi}{\partial x^i}. \tag{21}$$

The fit is the chain rule through the particle map. It is exact for any arrangement, any weights and any metric, provided the full M_a^{-1} is kept.

V. OPERATION 2: QUADRATIC FORMS

The second operation is where the metric matters. A quadratic form in two gradients is built as a bond sum, and the question is which metric that sum actually uses.

Let ϕ and χ be two fields, not necessarily linear. Split each bond difference into the part the fit captures and a residual:

$$\Delta\phi_{ab} = \nabla_i\phi\, r_{ab}^i + \delta_{ab}(\phi), \tag{22}$$

and similarly for χ. Because $\nabla_i\phi$ is the least-squares fit, the residual is orthogonal to the bond vectors:

$$\sum_{b\neq a} V_b\,\psi_{ab}\,\delta_{ab}(\phi)\,r_{ab}^j = \tilde{\nabla}^j\phi - \nabla_i\phi\,M_a^{ij} = 0. \tag{23}$$

When the bond sum of the products of differences is expanded, the cross terms therefore vanish. This leaves the exact identity

$$\sum_{b\neq a} V_b\,\psi_{ab}\,\Delta\phi_{ab}\,\Delta\chi_{ab} = \nabla_i\phi\;M_a^{ij}\;\nabla_j\chi + \sum_{b\neq a} V_b\,\psi_{ab}\,\delta_{ab}(\phi)\,\delta_{ab}(\chi). \tag{24}$$

It holds for any arrangement, any weights and any fields. For $\phi = \chi$ the residual term is non-negative. The first term is the *resolved* part and the second the *residual*. Three remarks on the size of the residual:

- *Smooth fields.* For a field that varies on a scale $L \gg h$, the residual is the curvature of the field across the star,

  $$\delta_{ab}(\phi) \simeq \tfrac{1}{2}\,\partial_i\partial_j\phi\;r_{ab}^i r_{ab}^j - \big(\text{its fitted linear part}\big). \tag{25}$$

  The residual term in eq. (24) is then smaller than the resolved term by a factor of order h^2/L^2. Because the cross terms vanish exactly, the correction is second order in the curvature, not first.

- *Fit versus true gradient.* The identity is written in terms of the fit $\nabla_i\phi$, so it contains no error from the fit itself. The fit differs from the true $\partial_i\phi$ by $O(h^2\,\partial^3\phi)$ on a symmetric star and by $O(h\,\partial^2\phi)$ on an irregular one. For an exactly linear field the residual vanishes and the fit is the true gradient, so the identity holds with no correction.

- *Unresolved modes.* For wavelengths comparable to the particle spacing, such as the zigzag mode, the fit captures little and the residual is not small. It is the residual that damps these modes, see eq. (51) of Ref. [1].

The resolved term contracts the two gradient indices with M_a. The physics wants them contracted with h^{ij}. The resolved part is therefore the physical quadratic form only if

$$M_a^{ij} = n_a\, h^{ij}, \tag{26}$$

that is $M_a = n_a \mathbb{1}$ at rest and $M_a = n_a \ell_a$ for a moving fluid. For a vector field such as b, each pair of components β^k, β^l enters eq. (24), and the component indices are contracted separately, again with h_{kl} or h^{kl} as their position requires.

VI. NORMALIZED WEIGHTS: FACES, GAUSS AND THE LAPLACIAN

When the target eq. (26) holds, the factor n_a can be absorbed into the weights. The resulting normalized weights have a simple geometric meaning, and they connect the bond construction to finite volumes and to graph Laplacians.

Define

$$w_{ab} = \frac{V_b\, \psi_{ab}}{n_a}, \qquad \sum_{b\neq a} w_{ab}\, r_{ab}^i r_{ab}^j = h^{ij}. \tag{27}$$

The second equation is eq. (26). The weights have units of $1/\text{length}^2$, and they belong to particle a: $w_{ab} \neq w_{ba}$ in general. We use $V_b\psi_{ab}$ in Operations 1 and 2, where M_a is arbitrary, and w_{ab} once eq. (26) is imposed (with the corrected weights of Sect. IX it holds exactly).

a. Faces. Define the face vector of the bond,

$$A_{ab}^i = V_a\, w_{ab}\, r_{ab}^i, \qquad \sum_{b\neq a} A_{ab}^i\, r_{ab}^j = V_a\, h^{ij}. \tag{28}$$

A_{ab}^i has units of area and points along the bond. Its normal covector,

$$A_{ab\,i} = h_{ij}\, A_{ab}^j, \qquad \sum_{b\neq a} A_{ab\,i}\, r_{ab}^j = V_a\, \delta_i{}^j, \tag{29}$$

makes the face orthogonal to the bond in the metric h, i.e. in the rest frame of a. Then

$$V_a\, w_{ab} = \frac{|\boldsymbol{A}_{ab}|}{|\boldsymbol{r}_{ab}|}, \tag{30}$$

the face area over the distance between the particles. In finite volumes this ratio is the transmissibility of the face. The closure is the discrete form of $\oint x^j\, d\Sigma_i = V\delta_i{}^j$, with the factor of two from pairing the face with the full bond rather than the half-bond to its centre absorbed into A_{ab}. A momentum flux Π^{ij} at a gives the momentum $\Pi^{ij}A_{ab\,j}$ to b through the face. This quantity (known as the traction) is the momentum given by a to b along the bond ab.[1] The closure recovers the flux from its faces,

$$\sum_{b\neq a} \Pi^{ik} A_{ab\,k}\, r_{ab}^j = V_a\, \Pi^{ij}. \tag{31}$$

[1] In much of the literature on elasticity the traction is minus this quantity.

b. Gauss. With eq. (27) the fit eq. (18) becomes

$$\nabla_i \phi = h_{ij} \sum_{b \neq a} w_{ab}\, \Delta\phi_{ab}\, r^j_{ab} = \frac{1}{V_a} \sum_{b \neq a} \Delta\phi_{ab}\, A_{ab\,i}. \tag{32}$$

This is the finite-volume gradient: the flux of ϕ through the faces divided by the volume. It is written with differences, and the closure makes it exact for linear fields. The moment-weighted gradient is now $\tilde{\nabla}^i \phi = n_a\, h^{ij} \nabla_j \phi$.

c. Graph Laplacian. For a smooth field on a symmetric star the first moment $\sum_b w_{ab} r^i_{ab}$ vanishes, and expanding $\Delta\phi_{ab} = r^i \partial_i \phi + \frac{1}{2} r^i r^j \partial_i \partial_j \phi$ gives

$$\sum_{b \neq a} w_{ab}\, (\phi_b - \phi_a) \simeq \frac{1}{2}\, h^{ij} \partial_i \partial_j \phi. \tag{33}$$

So $2w_{ab}$ are the edge weights of a consistently normalized graph Laplacian, in the metric h. Its quadratic form is the bond sum of Operation 2,

$$E[\phi] = \frac{1}{2} \sum_a V_a \sum_{b \neq a} w_{ab}\, (\Delta\phi_{ab})^2 = \frac{1}{2} \sum_a V_a \big[h^{ij} \nabla_i \phi\, \nabla_j \phi + \text{residual} \big] \;\longrightarrow\; \frac{1}{2} \int h^{ij}\, \partial_i \phi\, \partial_j \phi, \tag{34}$$

by eq. (24). Its derivative is a sum over bonds with the symmetric weights $c_{ab} = V_a w_{ab} + V_b w_{ba}$,

$$-\frac{\partial E}{\partial \phi_a} = \sum_{b \neq a} c_{ab}\, (\phi_b - \phi_a) \simeq V_a\, h^{ij} \partial_i \partial_j \phi, \tag{35}$$

the graph Laplacian on the bonds. It has $\phi = \text{const}$ as its only zero mode when the bonds connect all particles, and every bond contributes a non-negative amount to E.

d. On a grid. For nearest neighbours at spacing Δx at rest, eq. (27) gives $w_{ab} = 1/(2\Delta x^2)$, so $V_a w_{ab} = \Delta x^{d_s - 1}/(2\Delta x)$, half the face area over the spacing. The bond Laplacian is then $\sum_b c_{ab}(\phi_b - \phi_a) = V_a \sum_b (\phi_b - \phi_a)/\Delta x^2$, the standard finite-difference Laplacian times the cell volume.

VII. WHAT SETS THE BOND MOMENT

Operation 2 requires the bond moment to equal the inverse metric, up to normalization. Here we collect what determines the bond moment, and so whether the requirement can hold.

a. The shape of the weights. Weights that are round in a metric give a moment proportional to the inverse of that metric, eq. (14). This is why ψ is evaluated at the comoving distance: it makes the continuum moment satisfy eq. (26), with $n_a \simeq n_0/\sqrt{\det h_{ij}}$.

b. The arrangement does not enter. The weights must be attached to physical (comoving) distances. For contrast, consider *reference-lattice weights*: each bond keeps the weight it had on the reference lattice, $\psi(|\Delta\sigma_{ab}|)$, which is never updated as the particles move, as on a mesh. Then $\mu^{JK} = \mu_0 \delta^{JK}$ is fixed, and

$$M^{ij}_a = \mu_0\, e^i{}_J\, e^j{}_J, \tag{36}$$

the inverse of the metric $e^i{}_I e^i{}_J$ that the deformation induces on the reference lattice. The moment would record the deformation. With physical weights a smoothly deformed lattice is still a uniform point set, and the deformation drops out of M_a. This includes the Lorentz contraction: a lattice uniform in the fluid rest frame is a contracted lattice in the lab, and the weights $\psi(\rho)$ still give $M_a \propto h^{ij}$. The metric in the weights sets the moment; the arrangement does not.

c. The sampling. A finite, irregular set of neighbours gives M_a a sampling error. With physical weights the sampling error is the only source of anisotropy. It is removed by the correction Λ_a below.

So the fit never needs eq. (26), while the quadratic form does. The mean-field form of Sect. IV C of Ref. [1] uses both operations. Operation 1 gives Ω_a and θ_a. The covariant bond differences $\mathcal{D}_{ab}\mathsf{b}$ of eq. (44) of Ref. [1] then enter the quadratic form of Operation 2. If eq. (26) fails, directions that happen to be over-represented among the neighbours are over-weighted in K of eq. (45) of Ref. [1].

VIII. THE SAMPLING ERROR

The previous section left sampling error as the only departure from the target. We now measure its size, first at rest and then in motion. The measure is the anisotropy of the moment relative to its target,

$$A_a = \frac{\left|\hat{M}_a - \frac{1}{d_s}\operatorname{tr}\hat{M}_a\,\mathbb{1}\right|}{\frac{1}{d_s}\operatorname{tr}\hat{M}_a}, \qquad \hat{M}_a = h^{1/2} M_a h^{1/2}, \tag{37}$$

which vanishes exactly when eq. (26) holds. `examples/moment_sampling.jl` measures it at rest for a perfect hexagonal lattice, jittered lattices (each particle shifted uniformly in $\pm ja/2$ per coordinate, the convention of `setup_ic`), random (Poisson) particles, and a lattice under a smooth 10% shear, with physical and, for contrast, reference-lattice weights (Fig. 2, Table I).

- The perfect lattice gives $A_a = 0$. The smooth shear gives $A_a \sim 10^{-4}$ with physical weights, falling rapidly with h/a; with reference-lattice weights it gives $A_a = 0.26$ at every resolution.

- Jitter gives a genuine sampling error, linear in the amplitude and falling as $(a/h)^2$:

$$\overline{A}_a \simeq 0.4\, j \left(\frac{a}{h}\right)^2, \tag{38}$$

  with the maximum about three times the mean. Each of the $N_{\mathrm{nb}} \sim (h/a)^2$ neighbours shifts the moment by a random relative amount $\sim (ja/h)/N_{\mathrm{nb}}$; in quadrature these give $\sim j(a/h)^2$.

- Poisson particles converge only as $1/\sqrt{N_{\mathrm{nb}}} \sim a/h$, since each neighbour contributes an $O(1)$ random amount. At $h/a = 2.55$ the mean is 0.17 and the maximum 0.54. Poisson-like disorder is where Λ_a is most likely to fail.

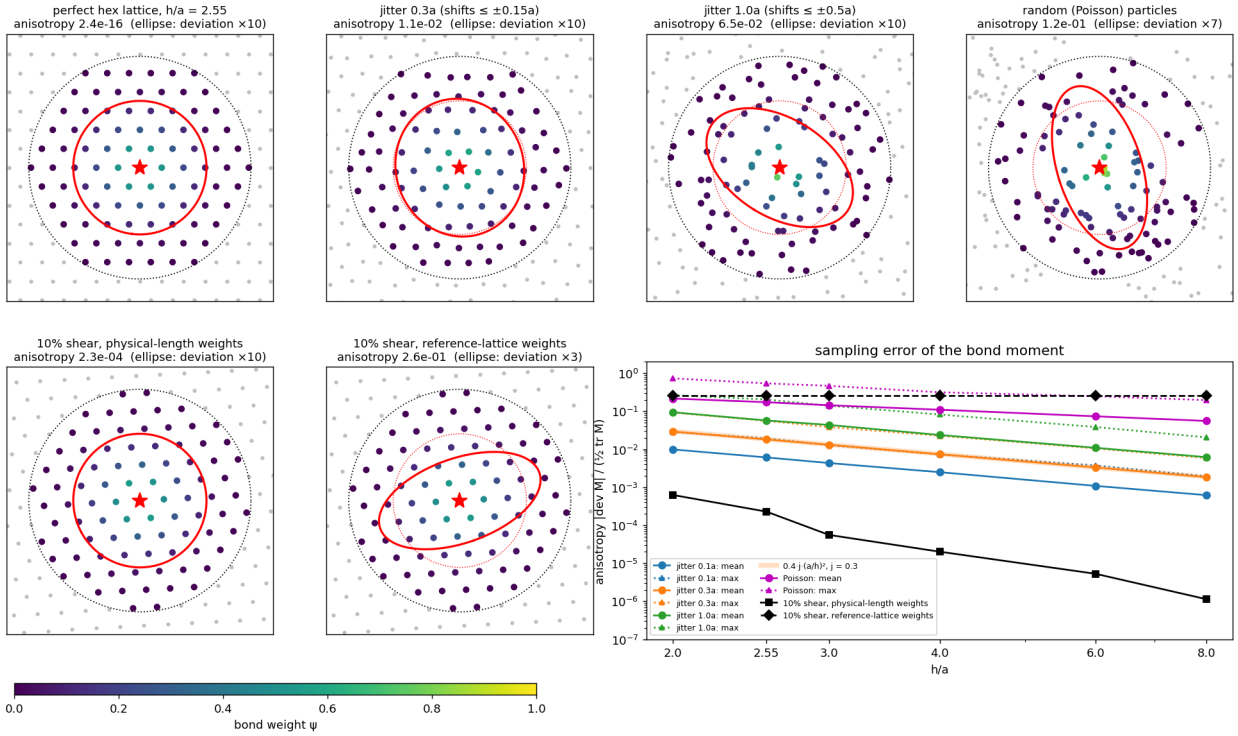


FIG. 2. The bond moment as a sampled quantity. Six star panels: the star of the particle marked by the red star at $h/a = 2.55$ (dots coloured by the bond weight ψ, kernel support dotted). The red ellipse is the normalized moment $M_a/(\frac{1}{2}\,\mathrm{tr}\,M_a)$, with its deviation from the unit circle (red dotted) exaggerated by the factor shown. Bottom right: mean (circles) and maximum (triangles) anisotropy over 2000 random stars as a function of h/a. The curves are for jitters $0.1a$, $0.3a$, $1.0a$ (shifts uniform in $\pm ja/2$) and for Poisson particles, together with the smooth 10% shear for physical-length (black squares, the code) and reference-lattice (black diamonds, mesh-like: each bond keeps its weight from the undeformed lattice) weights. The shaded line is eq. (38) for $j = 0.3$.

a. Moving fluid. For a moving fluid the stencil $\rho < 2h_\psi$ covers a lab area smaller by $1/\gamma$, so it holds $1/\gamma$ as many neighbours, and the sampling error grows accordingly. For a jittered ($j = 0.3$) lab lattice, with 1000 random stars per entry and A_a computed with $\hat{M}_a$:

		$h/a = 2.55$			$h/a = 4$		
$\boldsymbol{v}$	γ	N_{nb}	mean	max	N_{nb}	mean	max
0	1	91	1.8×10^{-2}	5.1×10^{-2}	233	7.3×10^{-3}	2.3×10^{-2}
$0.5\,\hat{x}$	1.155	81	2.1×10^{-2}	7.0×10^{-2}	203	8.7×10^{-3}	2.8×10^{-2}
$0.8\,\hat{x}$	1.667	57	3.2×10^{-2}	9.2×10^{-2}	138	1.30×10^{-2}	3.8×10^{-2}
$0.8\,(\hat{x}+\hat{y})/\sqrt{2}$	1.667	54	3.2×10^{-2}	1.0×10^{-1}	137	1.32×10^{-2}	3.8×10^{-2}

The direction of $\boldsymbol{v}$ relative to the lattice does not matter, and the mean grows roughly as γ (by 1.78 at $\gamma = 1.67$). eq. (38) becomes

$$\overline{A}_a \simeq 0.4\, j\, \gamma \left(\frac{a}{h}\right)^2. \tag{39}$$

The effective resolution is set by the rest-frame spacing, not the lab spacing.

		jitter $0.1a$		jitter $0.3a$		jitter $1.0a$		Poisson		10% shear	
h/a	$N_{\rm nb}$	mean	max	mean	max	mean	max	mean	max	phys.	ref.
2.00	58	9.9e-3	2.9e-2	2.9e-2	9.3e-2	9.4e-2	0.27	0.22	0.73	6.3e-4	0.26
2.55	90	6.1e-3	1.9e-2	1.8e-2	5.6e-2	5.7e-2	0.20	0.17	0.54	2.3e-4	0.26
3.00	124	4.3e-3	1.3e-2	1.3e-2	3.8e-2	4.4e-2	0.14	0.14	0.47	5.6e-5	0.26
4.00	238	2.5e-3	7.3e-3	7.4e-3	2.3e-2	2.4e-2	8.2e-2	0.11	0.32	2.0e-5	0.26
6.00	514	1.1e-3	3.8e-3	3.3e-3	1.0e-2	1.1e-2	3.9e-2	7.4e-2	0.25	5.3e-6	0.26
8.00	928	6.2e-4	1.9e-3	1.9e-3	5.9e-3	6.2e-3	2.1e-2	5.6e-2	0.20	1.1e-6	0.26

TABLE I. Anisotropy A_a of the bond moment (2000 random stars per entry). $N_{\rm nb}$ is the number of neighbours inside the support $2h$ on the lattice. Jitter j: shifts uniform in $\pm ja/2$ per coordinate. Shear: physical-length weights (phys., the code) and reference-lattice weights (ref.). Between $h/a = 2.55$ and 8 the means fall as $(h/a)^{-2.00}$ (jitter $0.3a$), $(h/a)^{-1.94}$ (jitter $1.0a$) and $(h/a)^{-0.99}$ (Poisson).

IX. SAMPLING CORRECTIONS: Λ_a

The sampling error is small but not negligible at the resolutions used in practice. Rather than accept it, we remove it particle by particle with a small change of the bond weights.

We reweight the bonds of each particle so that eq. (26) holds exactly. The corrected weight is

$$\psi_{ab} \;\to\; \psi_{ab}\big(1 + \Lambda_{a,kl}\, r^k_{ab} r^l_{ab}\big), \tag{40}$$

with $\Lambda_{a,kl}$ symmetric and carrying lower indices. Together with the number n_a, it is defined by

$$\boxed{\sum_{b\neq a} V_b\, \psi_{ab}\big(1 + \Lambda_{a,kl}\, r^k_{ab} r^l_{ab}\big)\, r^i_{ab} r^j_{ab} = n_a\, h^{ij}, \qquad h^{kl}\Lambda_{a,kl} = 0.} \tag{41}$$

The first relation is $d_s(d_s+1)/2$ equations, one for each component of a symmetric matrix; the second is one more. Together they form a linear system for the $d_s(d_s+1)/2$ components of Λ_a and n_a. The trace condition fixes the remaining freedom: a part $\Lambda_{a,kl} \propto h_{kl}$ would multiply each weight by $1 + c\rho^2$, changing the radial profile of ψ rather than the shape of the moment, and only rescaling n_a. Taking the h-trace of the first relation gives

$$n_a = \frac{1}{d_s} \sum_{b\neq a} V_b\, \psi_{ab}\big(1 + \Lambda_{a,kl}\, r^k_{ab} r^l_{ab}\big)\, \rho^2_{ab}. \tag{42}$$

a. Relation to inverting M_a. To first order in the anisotropy, work in the coordinates $\boldsymbol{s} = h^{1/2}\boldsymbol{r}$, where the target is $\mathbb{1}$. The fourth moment of an isotropic star is

$$\sum_b c_b\, s_i s_j s_k s_l = \frac{\mu_4}{d_s(d_s+2)}\big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big), \qquad c_b = V_b\psi_{ab}, \quad \mu_4 = \sum_b c_b\, \rho^4_{ab}. \tag{43}$$

Transforming back,

$$\Lambda_{a,ij} \simeq -\frac{d_s(d_s+2)}{2\mu_4}\Big(h_{ik} M_a^{kl} h_{lj} - \frac{1}{d_s}\big(h_{kl} M_a^{kl}\big)\, h_{ij}\Big). \tag{44}$$

At rest in two dimensions the prefactor is $4/\mu_4$. Bonds along over-represented directions are down-weighted in proportion to the deviator of M_a relative to h^{ij}. The reweighting is the weight-space version of acting on the moment with $n_a\, h^{il}(M_a^{-1})_{lk}$: it removes the shape of M_a and leaves its size to n_a.

b. At rest in two dimensions. Write $\boldsymbol{r}_{ab} = (x, y)$. A traceless Λ_a has two components,

$$\boldsymbol{r}^T \Lambda_a \boldsymbol{r} = \Lambda_a^{xx}(x^2 - y^2) + 2\Lambda_a^{xy} xy, \tag{45}$$

and the conditions $M_a^{xx} = M_a^{yy}$ and $M_a^{xy} = 0$ are linear in them:

$$\begin{pmatrix} \sum_b c_b (x^2-y^2)^2 & 2\sum_b c_b\, xy(x^2-y^2) \\ \sum_b c_b\, xy(x^2-y^2) & 2\sum_b c_b\, x^2y^2 \end{pmatrix} \begin{pmatrix} \Lambda_a^{xx} \\ \Lambda_a^{xy} \end{pmatrix} = -\begin{pmatrix} \sum_b c_b (x^2-y^2) \\ \sum_b c_b\, xy \end{pmatrix}, \tag{46}$$

with n_a then given by the trace. The right-hand side is the sampling anisotropy, so $\Lambda_a = 0$ on the perfect lattice; eq. (44) agrees with the exact solution to a few percent on jittered stars at $h/a = 2.55$.

c. Moving fluid. Here $h^{ij} = \ell_a^{ij}$ and the weights are $\psi(\rho_{ab})$ with $\rho_{ab}^2 = r^T \ell_a^{-1} r$. Equation (41) then reads

$$\sum_{b\neq a} V_b\, \psi(\rho_{ab}) \big(1 + \boldsymbol{r}_{ab}^T \Lambda_a \boldsymbol{r}_{ab}\big)\, \boldsymbol{r}_{ab}\boldsymbol{r}_{ab}^T = n_a\, \ell_a, \qquad \ell_a^{kl}\Lambda_{a,kl} = 0. \tag{47}$$

In 2D this is a 4×4 linear system for the three components of the symmetric Λ_a, constrained by the trace condition, and n_a. At rest Λ_a also had three components, but there we enforced the trace condition $\Lambda_a^{xx} + \Lambda_a^{yy} = 0$ directly. The 4×4 system is what the code solves (`bond_weights_comoving` in `src/celllist.jl`; the at-rest case is the same routine with $\ell_a = \mathbb{1}$). Three points:

- Each ingredient has a role. $\psi(\rho)$ makes the continuum moment comoving, eq. (14). Λ_a only removes the sampling error, and it is traceless with respect to ℓ_a, not $\mathbb{1}$. Without $\psi(\rho)$, Λ_a would have to absorb the $O(v^2)$ anisotropy $\ell_a - \mathbb{1}$ itself, which does not vanish as $h/a \to \infty$.
- The normalization is $n_a \simeq n_0/\gamma_a$, with corrections of the size of the sampling error.
- In one dimension the moment is a number, a traceless Λ_a vanishes, and no correction is needed; the moment itself fixes n_a.

The corrected weights are used everywhere: in the fit, in K of eq. (45) of Ref. [1], and in n_a.

X. THE LIMIT $h/a \to \infty$

The correction Λ_a is a discretization device, and it should disappear in the continuum. We check that it does.

What enters the weights is the dimensionless $\boldsymbol{r}^T\Lambda_a\boldsymbol{r}$ with $\rho < 2h$, and by eq. (44) it is of the order of the anisotropy A_a. Hence

$$\big|\boldsymbol{r}^T\Lambda_a\boldsymbol{r}\big| \;\lesssim\; O\Big(j\,\gamma\,\frac{a^2}{h^2}\Big) + O\Big(\frac{h^2}{L^2}\Big) \;\longrightarrow\; 0 \qquad (a \ll h \ll L). \tag{48}$$

The first term is the sampling error eq. (38) for disorder bounded at the particle scale; for Poisson-like disorder it is replaced by $O(a/h)$. The second term comes from an anisotropic curvature of the density on a scale L (the volume weights $V_b = s_b/S_b$ compensate the density itself, and a linear gradient cancels by symmetry). So Λ_a is a pure discretization correction, like a kernel renormalization, and disappears in the continuum. The exception is a boundary, where the star is one-sided and M_a is anisotropic at $O(1)$; in the periodic box there are no boundaries.

XI. WHEN THE REWEIGHTING FAILS

Finally, the linear system for Λ_a can produce negative weights when a neighbourhood is very poorly sampled.

If the neighbourhood is sampled too poorly, a corrected weight $1 + \boldsymbol{r}^T \Lambda_a \boldsymbol{r}$ can turn negative. The code then falls back to $\Lambda_a = 0$, with n_a from the trace of the uncorrected moment. The fit remains exact, but for that particle the resolved part of eq. (24) is $\nabla_i\phi\, M_a^{ij}\, \nabla_j\phi$ rather than the physical $n_a\, h^{ij}\, \nabla_i\phi\, \nabla_j\phi$.

XII. SUMMARY

Gradients and quadratic forms on particles are both governed by the bond moment, the push-forward of the particles' metric to physical space. The fit of a gradient is exact on any arrangement. The bond sum is the physical quadratic form when the bond moment equals the inverse metric of the fluid. Bond weights evaluated at the comoving distance achieve this for smoothly distributed particles, at rest and in motion, up to a sampling error that falls as $(a/h)^2$ for bounded disorder. The per-particle correction Λ_a removes that error exactly and vanishes in the continuum limit.

---

[1] D. Teaney, "One-level (bond) dissipation on particles: one and two dimensions, at rest and in motion", notes (`bond_dissipation.tex`).